

CHALLENGE TRACK 1 · QUANTUM CHEMISTRY

Mapping the Energy Landscape of a Molecule

From a single ground-state energy to the physics of a chemical bond.

Background and Motivation

The energy of a molecule as a function of the positions of its nuclei — its **potential energy surface (PES)** — is one of the most useful objects in chemistry. Its minimum tells us the equilibrium geometry; its curvature encodes vibrational frequencies; the way it flattens as atoms separate describes how bonds break. Computing it accurately means solving the electronic structure problem repeatedly, and the classical cost of doing so grows steeply with system size.

Quantum computers offer a natural route: a molecule's electronic state can be encoded into qubits, and its energy estimated variationally. IBM Quantum's 2017 demonstration of a hardware-efficient variational quantum eigensolver (VQE) for small molecules — the paper that inspires this track — showed that real superconducting hardware can estimate molecular ground-state energies. A decade into the era of cloud-accessible quantum computers, this challenge invites you to go beyond a single energy and map out the physics of a bond.

> **The scientific question**
>
> How does the ground-state energy of a molecule change as we stretch, compress, and break its bonds — and how far can a quantum-computed potential energy surface be trusted as the system grows?

The problem

For a molecule with fixed nuclei, the electronic structure problem is to find the lowest eigenvalue E_0 of the electronic Hamiltonian $\hat{H}$, i.e. the ground-state energy. VQE approaches this through the variational principle: for any parameterized trial state $\left|\psi(\vec{\theta})\right\rangle$,

$$E(\vec{\theta}) \;=\; \frac{\left\langle \psi(\vec{\theta}) \right| \hat{H} \left| \psi(\vec{\theta}) \right\rangle}{\langle \psi(\vec{\theta}) | \psi(\vec{\theta}) \rangle} \;\geq\; E_0, \tag{1}$$

so minimizing $E(\vec{\theta})$ over the parameters approximates the ground state from above. A *potential energy surface* is obtained by repeating this minimization at a series of nuclear geometries and plotting E_0 against a geometric coordinate (e.g. a bond length).

Because the challenge is about *which molecule* and *which geometry*, each level below is specified by a target molecule rather than by an explicit Hamiltonian — you construct the Hamiltonian from the

molecule, basis set, and mapping of your choice. The companion tutorial notebook does this end to end for H_2 at its equilibrium bond length.

Getting Started — Beginner

Your target

Compute and plot the ground-state **potential energy surface of the HeH^+ molecular ion**: its total ground-state energy E_0 as a function of the He–H internuclear distance, across a range that captures both the equilibrium minimum and the onset of dissociation. Identify the equilibrium bond length from your curve.

Suggested milestones

These are optional pointers, not requirements — how you reach the target is up to you.

- Adapt the tutorial workflow to HeH^+ (mind the +1 charge and the electron count).
- Sweep the internuclear distance and solve for the ground state at each point.
- Locate the minimum of the curve and read off the equilibrium bond length.
- Comment on the shape near equilibrium and as the atoms are pulled apart.

Next Steps — Intermediate and Advanced

The two levels below are open-ended: the target defines *what* to achieve, and the milestones merely suggest *how*.

Intermediate — Do More With Less

Your target

Compute the ground-state energy of a larger molecule — **LiH or BeH_2** — to within chemical accuracy ($\approx 1.6 \times 10^{-3}$ Ha), while *reducing* the quantum resources required compared to a naive full-space UCCSD calculation. Report both the energy you reach and the resources (qubits, circuit depth, or measurements) you saved.

Suggested milestones

- Explore levers such as alternative fermion-to-qubit mappings, symmetry-based qubit tapering, freezing core orbitals, or grouping commuting measurements.
- Quantify the trade-off: report accuracy achieved against qubits, circuit depth, or shot budget.
- Decide which combination of techniques buys the most under a fixed budget, and explain why.

Advanced — Beyond a Single Ansatz

Your target

Estimate the ground-state energy of the **nitrogen molecule (N_2)** near its equilibrium bond length — a strongly correlated system where a naive UCCSD run struggles — using a sampling / beyond-ansatz approach that lets classical post-processing do the heavy lifting. Show that your result improves on what a plain variational ansatz achieves for the same system.

Suggested milestones

- Choose your strategy (e.g. sample-based quantum diagonalization) and justify why it suits a strongly correlated molecule like N_2.
- Compare your result against a plain UCCSD run on the same system to show the improvement.
- Validate correctness where exact diagonalization is infeasible, e.g. against a classical CI / CCSD reference.

Running on real hardware (all levels)

Moving from the statevector simulator to a real IBM Quantum device (such as `ibm_basquecountry`) changes the game: heavy-hex connectivity, finite shots, and gate noise all affect the energy you measure. Comparing ideal simulation, noisy simulation, and real-device runs, and applying error suppression / mitigation and hardware-aware transpilation, is valuable at every level.

Suggested Resources

Qiskit Nature (quantum chemistry)

Qiskit Nature documentation
Entry point for electronic-structure problems, drivers, mappers, and chemistry ansätze.

Ground-state solvers tutorial
Finding molecular ground states with VQE and classical solvers — closest to the tutorial notebook.

Qubit mappers tutorial
Jordan–Wigner, parity, and Bravyi–Kitaev mappings, plus symmetry-based qubit tapering — relevant to the intermediate level.

UCCSD ansatz documentation
Reference page for the UCCSD variational form used in the tutorial.

Variational algorithms, sampling, and optimizers

VQE documentation
Reference page for the Variational Quantum Eigensolver class.

AdaptVQE how-to
Builds an ansatz adaptively — a useful stepping stone toward the advanced level.

Sample-based quantum diagonalization (SQD) tutorial
Post-processing quantum samples to estimate the ground state of N_2 — directly relevant to the advanced target.

Hardware, transpilation, and error mitigation

Transpile with pass managers
Transpiling circuits for a specific backend and its connectivity.

Error mitigation and suppression techniques
Overview of the resilience options available through Qiskit Runtime.

Background reading

Kandala et al., Hardware-efficient VQE for small molecules and quantum magnets, Nature (2017)
The IBM Quantum paper that inspires this challenge track.